

# PHAM NGOC DUNG

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## PERSONAL PROFILE

I am a junior undergraduate student at University of Science, Ho Chi Minh City (VNU-HCM), with research interests in generative models and reinforcement learning. My focus spans reinforcement learning for large language models, including RLVR and GRPO-based post-training for reasoning and tool use, as well as flow-based policy learning for robotic control. I am applying to the GStar Bootcamp to deepen both my research and engineering expertise through rigorous training and mentorship from leading researchers.

## EDUCATION

### University of Science, VNU-HCM

*B.Sc. in Knowledge Engineering (3rd Year) — GPA: 3.8/4.0*

Ho Chi Minh City, Vietnam

*Sep. 2023 – Sep 2027 (Expected)*

- **Relevant coursework:** Linear Algebra, Probability & Statistics, Calculus, Data Science, Statistical Learning

## TECHNICAL EXPERIENCE

### Viettel Digital Talent Program — Data Science & AI

*Viettel Group — Selective 3-month technical training program*

Apr 2026 – Jul 2026

*Ho Chi Minh City, Vietnam*

- Selected as top 15% from a nationwide applicant pool of over 1200 candidates for the Data Science and Artificial Intelligence track; completed a 12-week curriculum spanning NLP to computer vision under faculty mentorship.
- Developed **ToolFormer** as the program capstone, implementing an end-to-end RLVR training pipeline for Qwen3-4B-Instruct on function-calling tasks.
- Ranked 36th of 180 teams/participants (top 20%) in the capstone project and invited to present the final solution before the program defense committee.

## RESEARCH EXPERIENCE

### Undergraduate Research Assistant

*Computational Linguistics Center, University of Science, VNU-HCM*

Oct 2025 – Feb 2026

*Ho Chi Minh City, Vietnam*

- Investigated efficient neural style transfer methods under Prof. Dinh Dien for synthetic Sino-Nom character image generation to improve OCR robustness.
- Proposed and prototyped FFT-based algorithms for representation disentanglement, achieving a 12% improvement in LPIPS over the baseline model.
- Presented findings and proposed methods at weekly lab seminars, incorporating graduate fellow feedback into subsequent experiment design.

### VinRobotics Research Internship

*VinRobotics Talent Programme*

Apr 2026 – Present

*Ho Chi Minh City, Vietnam*

- Conducting research on flow-based policy learning algorithms for robotics under the guidance of Senior Director Do V. Thach, with a focus on Flow Policy Optimization (FPO).
- Presented weekly experimental results, analyzed failure modes, and incorporated research feedback into subsequent algorithm iterations.
- Implemented and benchmarked recent policy learning methods against the FPO baseline on robotic manipulation tasks.

## SELECTED PROJECTS

### Tool Former | *PyTorch, transformers, unsloth*

**Code** | Apr 2026 – Jul 2026

- Fine-tuned Qwen3-4B-Instruct model on function calling tasks with RLVR methods, addressing the advantage collapse problem and achieving an 18× improvement on function selection accuracy compared to baseline.
- Conducted rigorous ablation studies on KL divergence, rollout group size and multiple GRPO loss variants for performance comparison.
- Discovered reasoning bias and designed reward formulations to prevent reward hacking.
- Integrated function retrieval modules for faster and more efficient inference pipeline, reducing inference latency by 30%.

**OuroTrace** | *PyTorch, transformers, lm-evaluation-harness*

**Code** | Oct 2025 – Dec 2025

- Benchmarked Ouro looped language models (ByteDance) against standard transformers on 5 reasoning tasks using **lm-evaluation-harness**.
- Built robust answer-extraction with custom prompts, stop sequences, and regex parsing, raising extraction accuracy for robust evaluation with reasoning benchmarks.
- Identified the optimal number of recurrent steps for latent reasoning, showing peak performance over baseline at acceptable computing cost, identifying the performance-compute trade-off for recurrent reasoning.

**FontDiffusion** | *PyTorch, diffusers, W&B*

**Code** | Oct 2025 – Jan 2026

- Built an end-to-end diffusion pipeline for font generation, from dataset construction (1000 glyphs across 20 fonts) to training with full experiment tracking in Weights & Biases.
- Extended FSTDiff with custom FFT-based modules that decouple content and style features, improving LPIPS by 12% over the baseline.
- Designed a content-invariance loss that improved generation quality on SSIM by 10%.
- Published datasets and model checkpoints to Hugging Face Hub for reproducibility.

## COMPETITIONS

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**NVIDIA Nemotron Model Reasoning Challenge** — Kaggle; 36/4182 teams

2026

**Thach Thuc Academic Competition** — 2nd Place Final

2026

## AWARDS & HONORS

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**Odon Vallet Scholarship** — Merit scholarship by Rencontres du Vietnam for outstanding academic achievement 2022, 2023

**Vietnam National Physics Olympiad** — Third Prize & Honorable Mention

2022, 2023

**30/4 Traditional Physics Olympiad (Southern Vietnam)** — Gold Medal, ranked 4th

2021

## SKILLS

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**Deep Learning:** PyTorch, Hugging Face Transformers, Diffusers, PEFT

**Reinforcement Learning:** PPO, SAC, Stable-Baselines3, Gymnasium

**Programming:** Python, C++

**Tools:** Git, Docker, Weights & Biases, JAX, Numba, Pandas

**Languages:** Vietnamese (native), English (fluent), German (beginner)